

High-Performance FFT Accelerator

1. Overview

This document tracks the HP team's exploration of FFT accelerator architectures targeting latency lower than 61.00 microseconds. It catalogues all explored directions — both completed and open — so the team can see what's been tried, what the results were, and where the remaining opportunities lie. Each team member is encouraged to claim open directions and add their own results directly into this document.

Current best result: Direction D3 achieves 170 cycles per N=32 FFT chunk (14.17 μ s at 12 MHz), a 4.31 $\times$ speedup over the 732-cycle baseline, with 15.6% less SoC area than D1/D2.

2. Exploration Directions

Green rows are completed and synthesis-verified. Open rows are available for the team. Claim a direction by writing your name in the Owner column. Add rows for new ideas.

#	Direction	Expected Impact	Status	Owner
D1	Register-file architecture	732 $\rightarrow$ 220 cycles (3.33 $\times$). Eliminates inter-stage SRAM round-trips.	☒ Verified (synth)	Shanghong Lin
D2	2 $\times$ parallel butterflies	220 $\rightarrow$ 206 cycles (3.55 $\times$). Two BF units + per-stage twiddle fill.	☒ Verified (synth)	Shanghong Lin
D3	SW twiddle preload (CSR)	206 $\rightarrow$ 170 cycles (4.31 $\times$). Removes S_LOAD_TWIDDLE + fill. -15.6% area.	☒ Verified (synth)	Shanghong Lin
D4	Preload input data (CSR)	Removes S_LOAD_DATA and S_STORE_DATA, 170 cycles $\rightarrow$ 42 cycles.	Parked	Shanghong Lin

D5	Wider memory bus / burst accelerator mem	Reduce LOAD/STORE from 128 to ~64 cycles (~36 cycles)	Open	
D6	Higher clock frequency (pipelining)	Linear speedup if BF critical path is split	☒ Verified (synth)	Shanghong Lin
D7	Radix 4 FFT		Open	
D8	Design spaces in physical design		Open	
D9			Open	

3. Direction Details

D1: Register-File Architecture

The baseline FFT accelerator performs all data storage in SRAM, requiring read–compute–write sequences per butterfly that span 9 FSM states. Approximately 89% of the 732-cycle runtime is spent on SRAM accesses, with only 11% on actual arithmetic.

D1 eliminates inter-stage SRAM round-trips by holding all 32 complex samples (64 words) in flip-flop registers. Data is loaded from SRAM once at the start, all 5 FFT stages execute from registers, and results are written back once at the end. This is the single largest optimisation: 732 → 220 cycles (3.33×).

Key trade-off: 2,368 flip-flops for the register file. Fits comfortably at 12 MHz but shows up in area reports. The accelerator accounts for 71.3% of total SoC area.

D2: 2× Parallel Butterflies

Two independent butterfly datapaths process pairs of butterflies simultaneously, halving the per-stage butterfly count. A per-stage twiddle fill sub-phase computes the full twiddle table via chained fixed-point multiplications of each stage’s primitive root. Total: 220 → 206 cycles (3.55×).

Key trade-off: The fill overhead (26 of 66 compute cycles) limits the benefit of parallelism. The second multiplier pair adds area but the critical path depth is unchanged.

D3: Software Twiddle Preload (CSR)

Firmware pre-computes the full $N/2 = 16$ global twiddle table and writes it into 32 CSR registers before asserting `enable_accel`. This eliminates both the `S_LOAD_TWIDDLE` FSM state and the per-stage fill sub-phase entirely. The FSM drops from 6 to 5 states, and `COMPUTE` executes pure butterflies from the first cycle of each stage. Total: 206 → 170 cycles (4.31×).

Key trade-off: Strictly better than D2 — faster (17.5% fewer cycles) and smaller (15.6% less SoC area). Twiddle accuracy also improves: single-shot quantisation from float → Q12 vs accumulated rounding from chained multiplies.

D4: Preload Input Data (CSR)

Firmware writes all 64 data words (32 real + 32 imaginary, bit-reversed) into CSR registers before asserting `enable_accel`, eliminating both `S_LOAD_DATA` and `S_STORE_DATA` from the FSM. The accelerator receives input data via a packed flat bus (identical to the twiddle preload mechanism in D3) and writes results back to CSR registers readable by the CPU. The FSM reduces from 5 states to 3: `INIT` → `COMPUTE` → `FINISH`. Total: 170 → 42 cycles (17.4×).

Key trade-off: The CPU-side cost is unchanged — each word still traverses the Flash → QSPI → PicoRV32 → iomem path, so total end-to-end latency improvement is limited to the 128 HW cycles saved. The CSR register count grows from 35 to 99 (adding 64 data words), and the packed flat bus to the FFT core becomes 2048 bits wide (vs 512 bits for twiddles alone), creating significant routing pressure in the ~75%-full core.

Alternative under consideration: Rather than eliminating `accelerator_mem` entirely, widening its interface (dual-port or multi-word reads) could halve or quarter the `LOAD/STORE` phases without the routing overhead of a full flat bus. Since `accelerator_mem` is a synthesised register file (not an SRAM macro), port count and width are freely configurable — only area and timing constrain the design space.

D6: 4-Stage Micro-Pipelined Compute Engine

The butterfly datapath is divided into four registered stages: `FETCH`, `MUL1`, `MUL2`, and `ADD`. This architecture pumps a new butterfly pair every clock cycle, achieving 1-throughput execution. For an $N=32$ chunk, each of the 5 FFT stages completes in 8 fetch cycles plus 3 drain cycles (11 cycles total), resulting in just 55 compute cycles. Stage advancement is triggered on the last drain cycle to eliminate dead bubbles between consecutive stages.

Key trade-off: Pipelining breaks up the long combinational path of the baseline, enabling the

maximum clock frequency to increase from ~12 MHz to ~60 MHz (a ~5× improvement). Combined with cycle count reductions, total latency per chunk drops from ~61 μs to ~3.1 μs (~20× faster). However, because compute is now highly optimized, memory bandwidth becomes the new binding constraint; LOAD and STORE phases now account for ~69% of the total 185 cycles.

5. Cross-Direction Comparison

Metric	Baseline	D1: Regfile	D2: +2× BF	D3: SW Preload	D6: Pipeline	D__:
Cycles / chunk	732	220	206	170	185	
Speedup vs baseline	1.00×	3.33×	3.55×	4.31×	3.96x	
Latency @ 12 MHz	61.0 μs	18.33 μs	17.17 μs	14.17 μs	At 60MHz: ~3.1μs	
Total SoC area	—	308,059	308,059	259,890		
Accel % of SoC	—	71.3%	71.3%	66.0%		
Worst-case slack	—	32.35 ns	32.35 ns	32.35 ns		
FSM states	—	6	6	5	5	
SRAM depth	—	128 words	128 words	64 words	64 words	

Empty columns are reserved for future directions.

6. Cycle Breakdown Analysis

Phase	Baseline	D2: +2× BF	D3: SW Preload	% of D3 Total
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INIT	2	1	1	0.6%
LOAD_TWIDDLE	—	10	0 (removed)	—
LOAD_DATA	~293	64	64	37.6%
COMPUTE	~730	66	40	23.5%
STORE_DATA	~293	64	64	37.6%
FINISH	1	1	1	0.6%
Total	732	206	170	100%

The dominant bottleneck has shifted: in the baseline, COMPUTE was ~89% of cycles. In D3, LOAD_DATA + STORE_DATA together are 75%. Any further HP optimisation must target the memory interface.